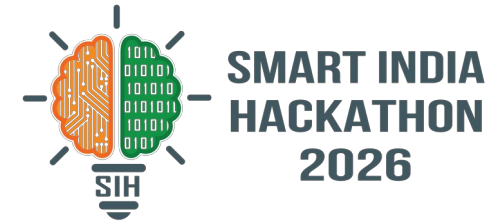
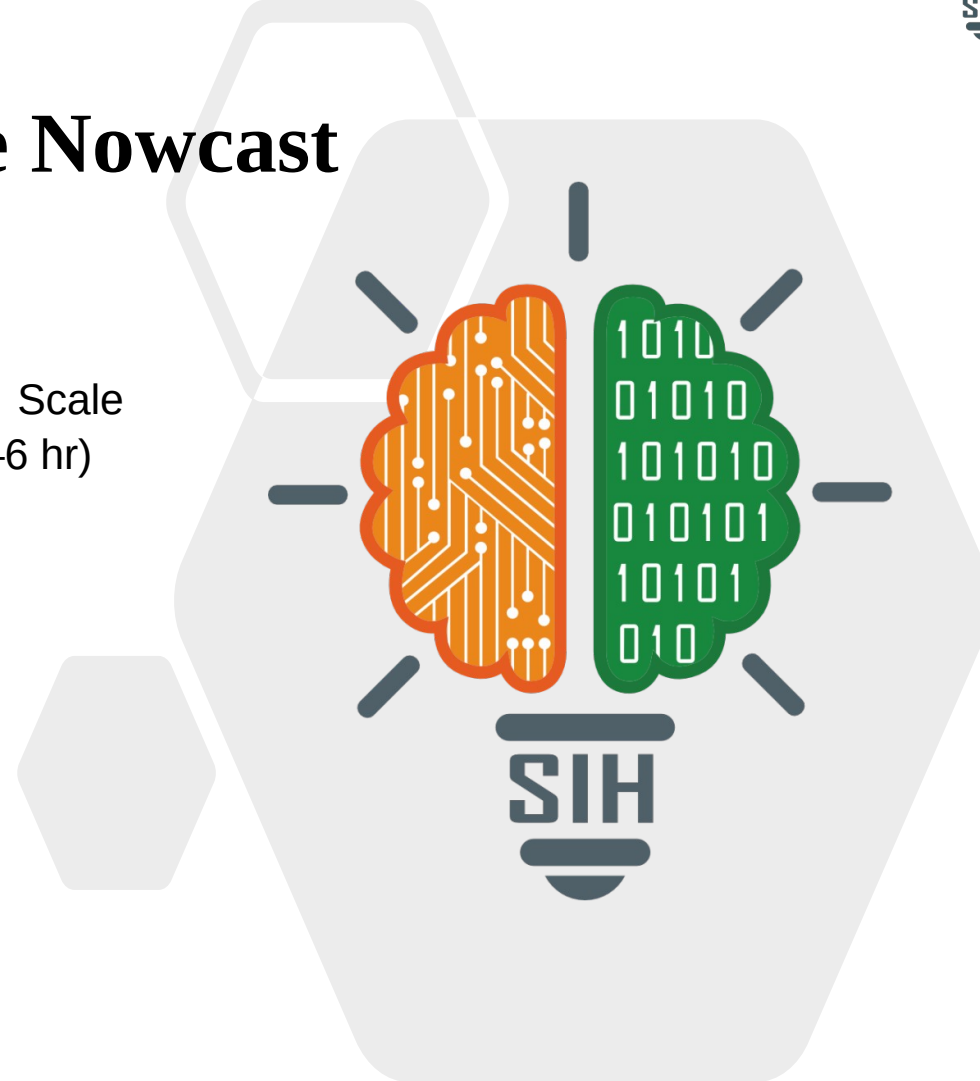


# SMART INDIA HACKATHON 2026



## StormLife Nowcast

- **Problem Statement ID** – SIH26084
- **Problem Statement Title-** Convective Scale  
Nowcasting for Thunderstorms, Hail & Cloudbursts (0–6 hr)
- **Theme-** Disaster Management
- **PS Category-** Software
- **Team ID-** 167858
- **Team Name** – THE FEVICONs

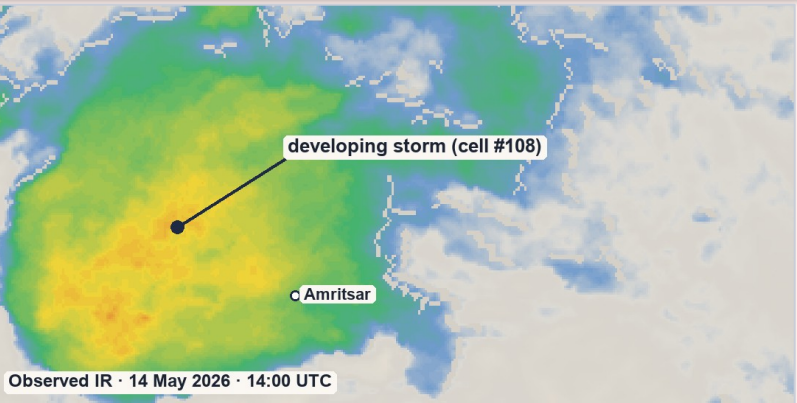


# IDEA TITLE

**StormLife Nowcast:** satellite → storm cell → a forecast you can check. Convective-scale nowcasting, 0–6 h: how far can it be trusted?

## ► HOW IT ADDRESSES THE PROBLEM

### WHAT WE SEE TODAY



? Where will it move?

? Will it intensify?

? How far can we trust the forecast?

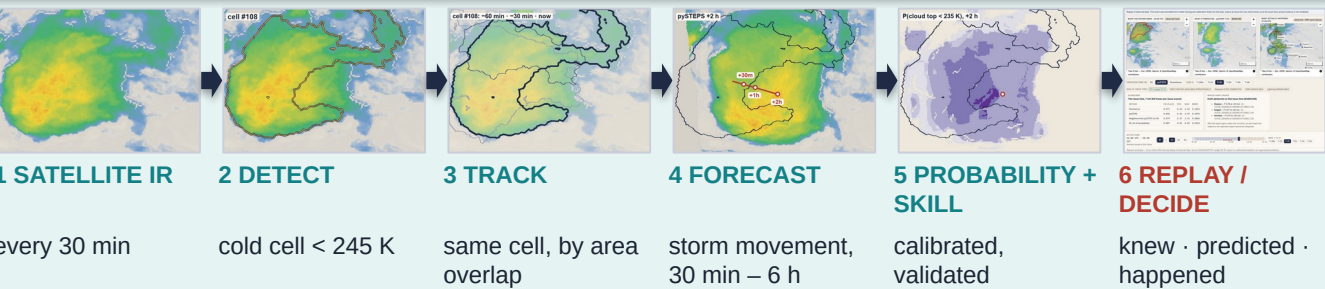
Thunderstorms, hail and cloudbursts grow from deep convection within hours · no open, live radar or lightning feed in India

30 min  
→ 6 h

2 km grid  
target: deep  
convection,  
cloud top < 235 K

## ► DETAILED EXPLANATION OF THE PROPOSED SOLUTION

### WHAT STORMLIFE ADDS satellite nowcasting of one real storm cell (#108, 14 May 2026)



**VALIDATED SKILL SHOWN  
BESIDE EVERY FORECAST**  
3 held-out storm days, pooled

**BSS 0.27**

ML probability skill at  
+2 h (no-fit ref. 0.02)

**to ≈ 4 h**

best probability skill of  
the 4 methods tested

**1 h**

yes/no maps: no gain  
over pySTEPS beyond  
1 h

## ► INNOVATION AND UNIQUENESS OF THE SOLUTION

**1 Real storm days only**  
No simulated radar, lightning or metrics;  
days fixed before scoring

**2 Baseline-first**  
Every result vs persistence and pySTEPS;  
negative results reported

**3 Calibrated probability**  
Reliability and validated skill shown with  
each forecast

**4 Audited and frozen**  
Tracking audited before use; 386 evidence  
files hash-locked

**RESEARCH PROTOTYPE** · real satellite data, held-out storm days · hazard-specific (hail, lightning, rain) skill not claimed · not live · not an official IMD warning

# TECHNICAL APPROACH

## ▶ TECHNOLOGIES & TOOLS

### 1 DATA & SCIENCE

Python  
NumPy  
pandas  
xarray  
SciPy

*Satellite processing & numerical analysis*

### 2 STORM INTELLIGENCE

OpenCV  
scikit-image  
custom overlap tracking  
optical flow

*Detect, track & characterize storm cells*

### 3 FORECAST & VALIDATION

pySTEPS  
scikit-learn  
isotonic  
calibration  
pytest

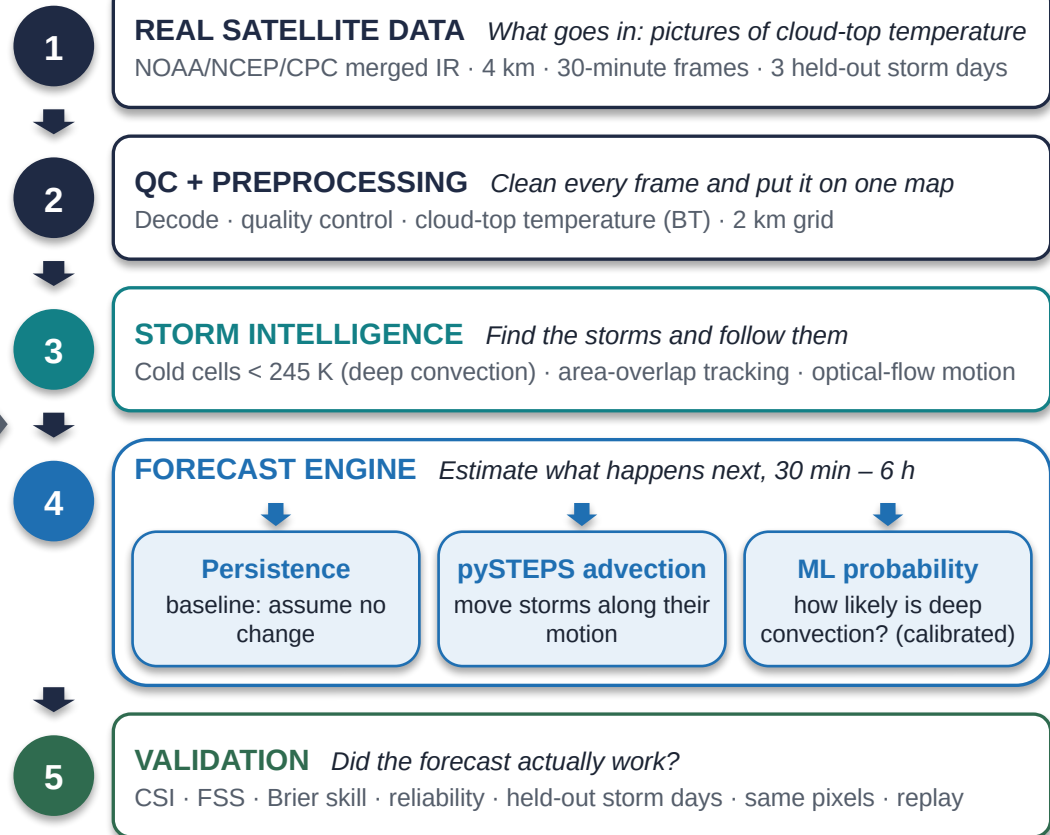
*Forecast, calibrate & verify*

### 4 PRODUCT

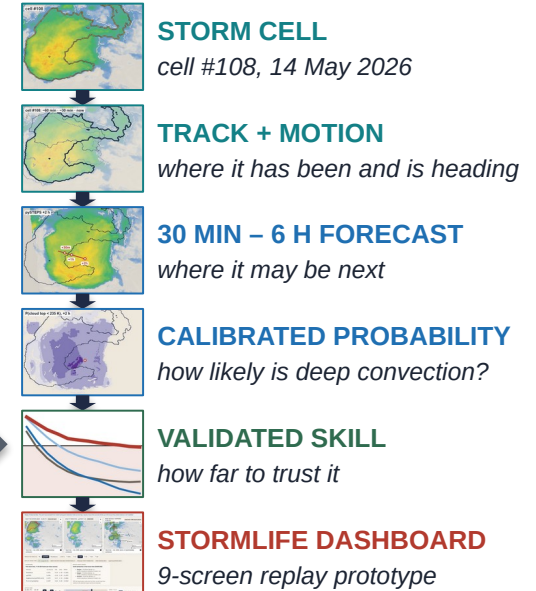
React  
TypeScript  
Vite  
MapLibre GL

*Interactive replay & decision-support dashboard*

## ▶ METHODOLOGY AND PROCESS: HOW STORMLIFE WORKS



## ▶ WORKING PROTOTYPE: WHAT IT PRODUCES



### STORMLIFE RESEARCH PROTOTYPE · replay, not live

Replay situation map

0–6 h forecast

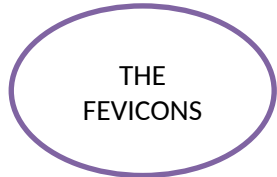
Historical replay

Model performance

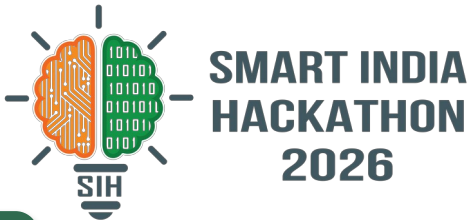
Data health

Exercise alert (CAP)

Real satellite data → audited storm intelligence → forecast + calibrated probability → validated decision support



# FEASIBILITY AND VIABILITY



► ANALYSIS OF THE FEASIBILITY OF THE IDEA: EVIDENCE AT A GLANCE

**25 days**  
600 real satellite files

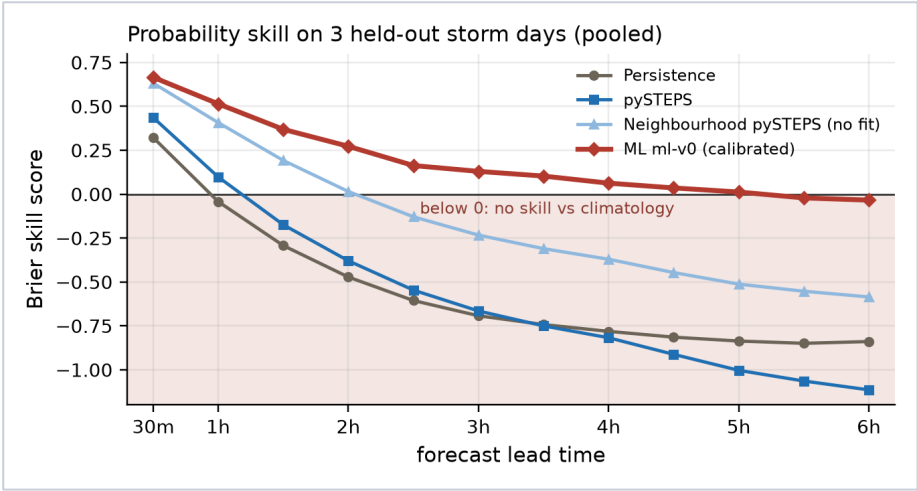
**3 storm days**  
held out: E8 · E10 · E11

**9 screens**  
working replay prototype

**84 tests pass**  
incl. 2 browser tests

**386 files**  
evidence hash-frozen

► WHAT THE TESTING SHOWED: 3 HELD-OUT STORM DAYS, POOLED, SAME PIXELS



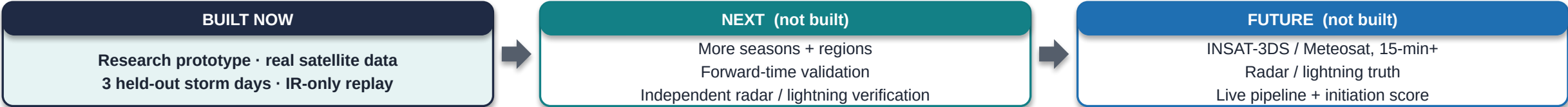
**3 / 3**  
**pySTEPS beats persistence**  
at every tested lead, 30 min – 6 h

**≈ 2 h**  
**Useful baseline skill**  
FSS 40 km useful to ~90 min (persistence) / ~120 min (pySTEPS)

**≈ 4 h**  
**ML probability skill**  
BSS 0.27 at 2 h (no-fit ref. 0.02) · calibration error 0.002–0.004

**1 h LIMIT**  
**LIMIT: no yes/no gain**  
ML does not improve binary yes/no maps over pySTEPS beyond 1 h

► POTENTIAL CHALLENGES AND RISKS → STRATEGIES FOR OVERCOMING THESE CHALLENGES: VIABILITY ROADMAP



Limits today: 3 held-out days (May 2026, NW India) · ML tested backward in time · single IR, 30-min · parallax not verified · no radar/lightning truth · replay, not live

# IMPACT AND BENEFITS

One storm signal → three decision users

**STORMLIFE: storm • forecast • skill • replay**

## ► POTENTIAL IMPACT ON THE TARGET AUDIENCE



## ► BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL, ETC.)

### SOCIAL

Potential earlier awareness of deep convection

### ECONOMIC

Open-data + open-source prototype

### ENVIRONMENTAL

Satellite-first; no new sensing hardware

### TRUST

Traceable metrics + visible limitations

*Research prototype • replay of real satellite data • not operational • exercise alerts only*

## RESEARCH AND REFERENCES

Research question → StormLife contribution → verified prototype → data sources → methods & reproducibility

## ► EXISTING APPROACHES

**RADAR**

cell tracking, e.g. MeteoSwiss  
TRT; IMD radar nowcasts

**LIGHTNING**

flash data in severe-storm  
guidance

**NWP**

model environment, e.g. in NOAA  
ProbSevere v3

**SATELLITE PRODUCTS**

storm objects + phases, e.g.  
NWCSAF RDT

Operational and research systems already provide storm detection,  
tracking and severe-weather guidance.

## ► OUR RESEARCH QUESTION

**WHAT CAN ACTUALLY BE  
VERIFIED FROM OPEN  
GEOSTATIONARY IR ALONE?**

## ► STORMLIFE CONTRIBUTION

**SATELLITE-FIRST**

open geostationary IR only

**BASELINE-FIRST**

vs persistence + pySTEPS

**CALIBRATED PROBABILITY**

reliability shown

**AUDITED TRACKING**

ID switches 21 % → 0.4 %

**REAL-EVENT REPLAY**

knew · predicted · happened

**TRACEABLE EVIDENCE**

every number → hashed file

**+ EXPLICIT LIMITATIONS** 3 days · single IR · no radar/lightning truth

## ► VERIFIED PROTOTYPE



3 held-out storm days • reproducible replay • validated metrics •  
evidence frozen

**StormLife complements existing operational systems; it  
does not replace them.**

Research prototype · real-event replay · held-out validation

## ► DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK: DATA SOURCES USED

**IMD**

Govt.  
of India

**India Meteorological Department**

**Used for:** hail-report dates to select storm days  
(context, not verification truth)

[Hail-storm report \(PDF\)](#) · [mausam.imd.gov.in](https://mausam.imd.gov.in)

**NOAA**

NCEP  
CPC

**NOAA / NCEP Climate Prediction Center**

**Used for:** Globally Merged IR: all satellite inputs  
+ truth · ~4 km · 30-min

[Merged IR page](#) · [Full-res IR](#) · [Archive used](#)



## ► METHODS &amp; REPRODUCIBILITY

**pySTEPS** → [pySTEPS official repo](#)  
external open-source: advection baseline + FSS

**STORMLIFE REPOSITORY** → [\[REPOSITORY URL\]](#)  
GitHub: our code · replay pipeline · validation · evidence

**DEMO / REPLAY** → [Demo video \[VIDEO LINK\]](#)  
short demo of the verified prototype